FragTree — Final Clustering Report

Generated: 2026-04-14 14:40:25 | min_cluster_size=75 | min_samples=53

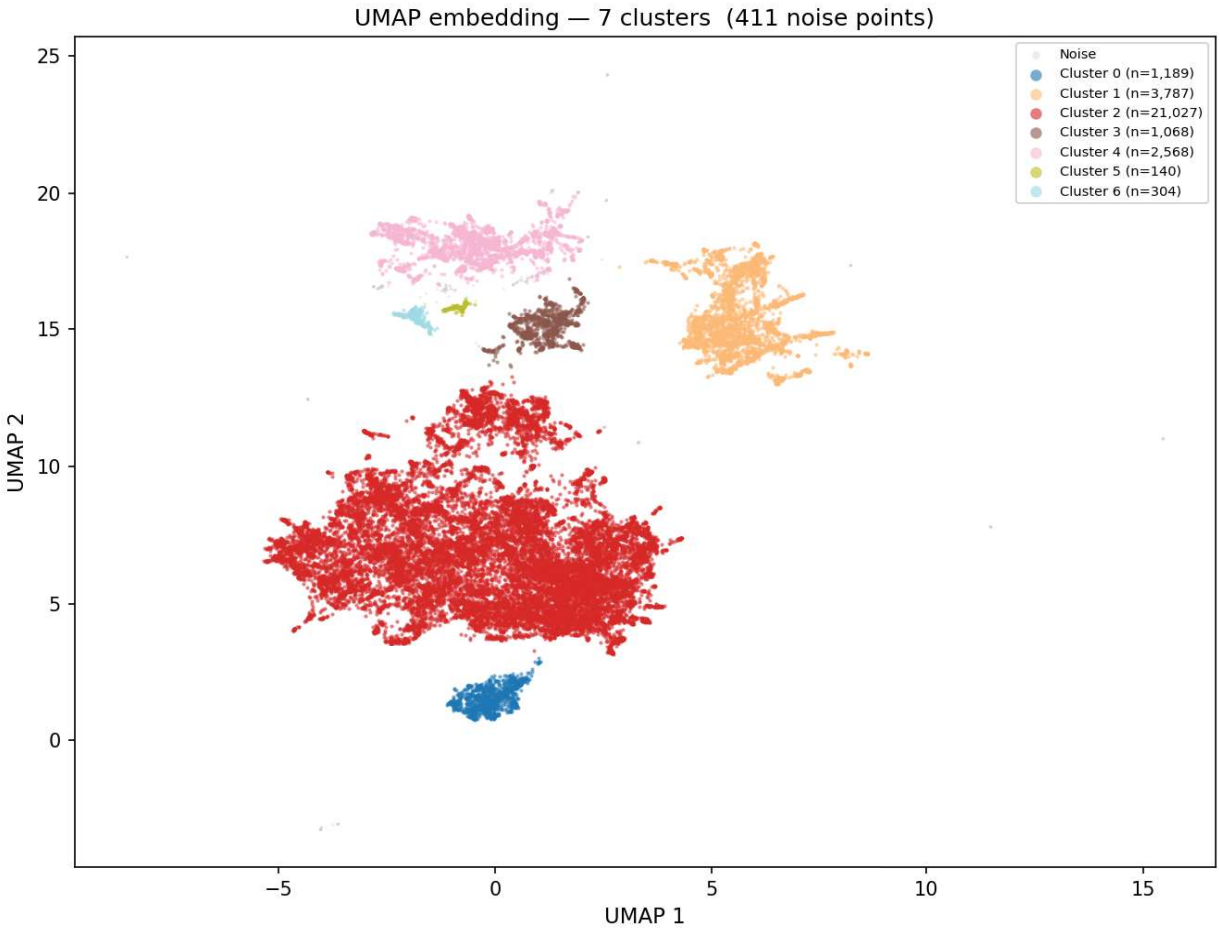
Global Clustering Metrics

DBCV is the primary metric. ARI and AMI use CAS number groupings as soft ground truth. BSS/TSS measures the fraction of total variance explained by the cluster structure (higher = better).

Metric	Value	Target	Notes
DBCV (primary)	0.2618	Maximize	Density-based clustering validation
Silhouette score	0.3115	Maximize	Mean intra vs inter-cluster cohesion
Noise fraction	0.013	< 0.30	411 points labelled noise
BSS / TSS	0.7300	Maximize	Fraction of variance explained by cluster structure
Mean persistence	0.2709	Maximize	Stability across density thresholds
N clusters	7	-	Noise cluster (-1) not counted
ARI	0.0000	1.0 ideal	Agreement with CAS number groupings
AMI	0.0000	1.0 ideal	ARI normalised for cluster size distribution

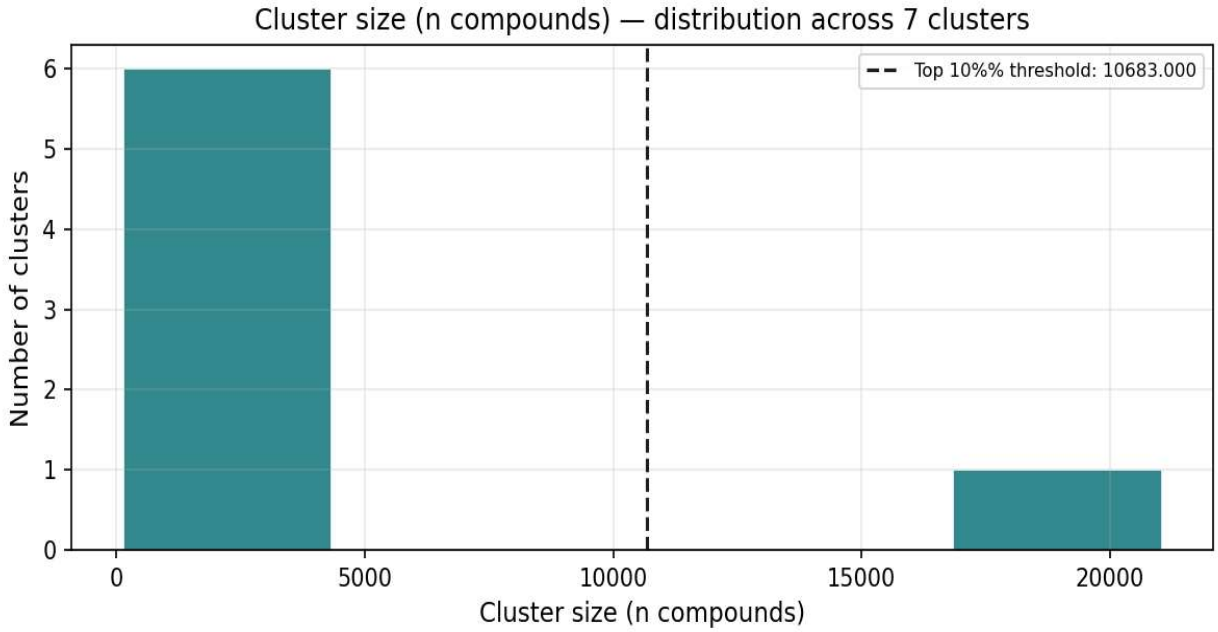
UMAP Embedding — Coloured by Cluster

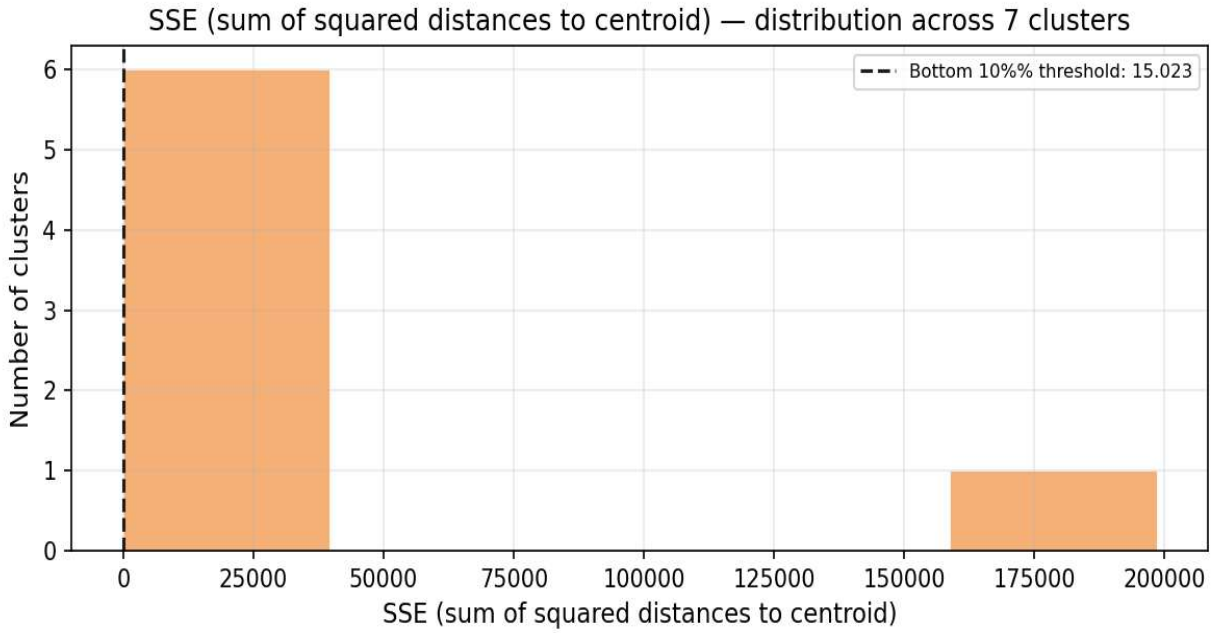
Each point is one compound. Grey = noise (label -1).

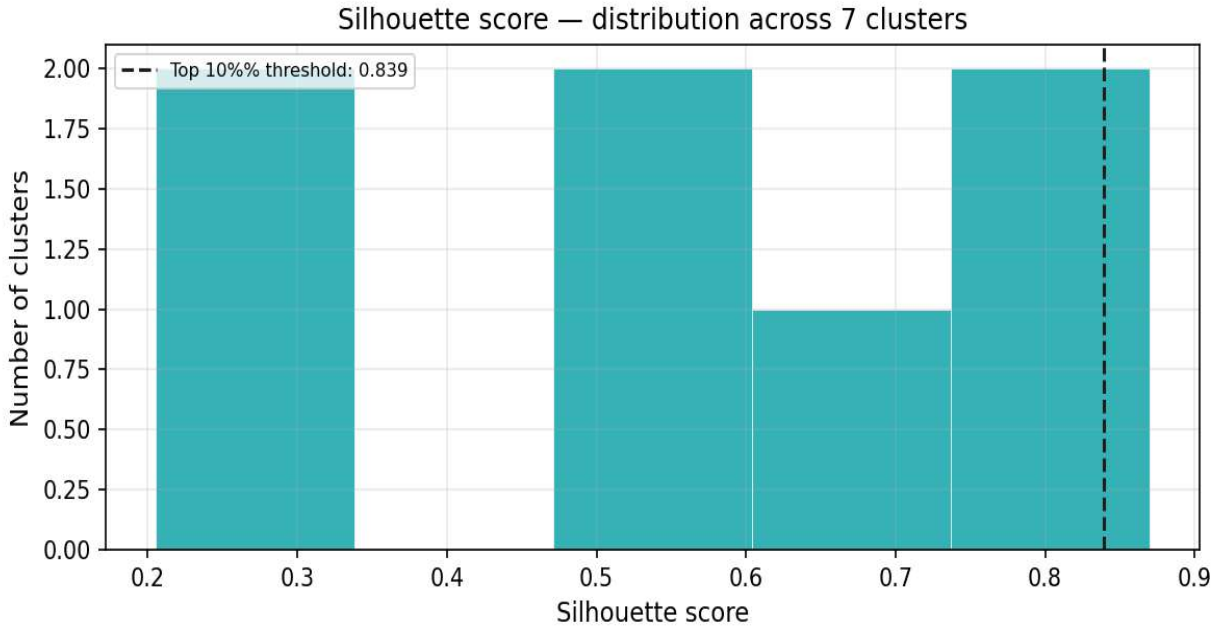

UMAP embedding — 7 clusters (411 noise points)
Noise
Cluster 0 (n=1,189)
Cluster 1 (n=3,787)
Cluster 2 (n=21,027)
Cluster 3 (n=1,068)
Cluster 4 (n=2,568)
Cluster 5 (n=140)
Cluster 6 (n=304)
UMAP 2
UMAP 1
25
20
15
10
5
0
−5
0
5
10
15

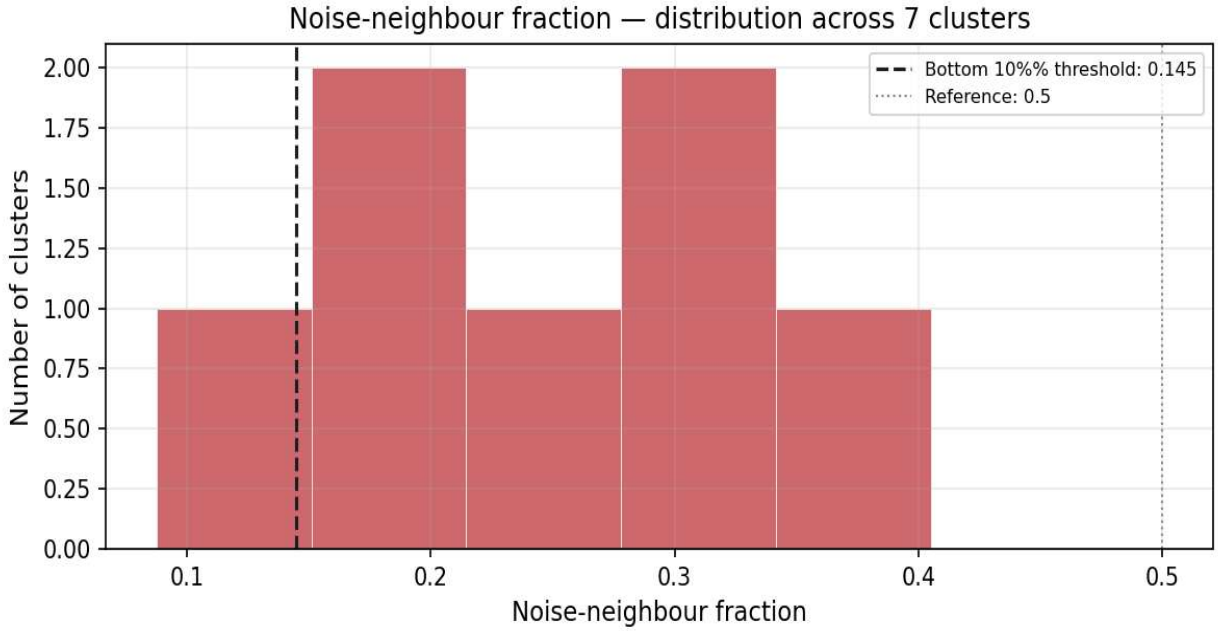
Per-Cluster Metric Distributions

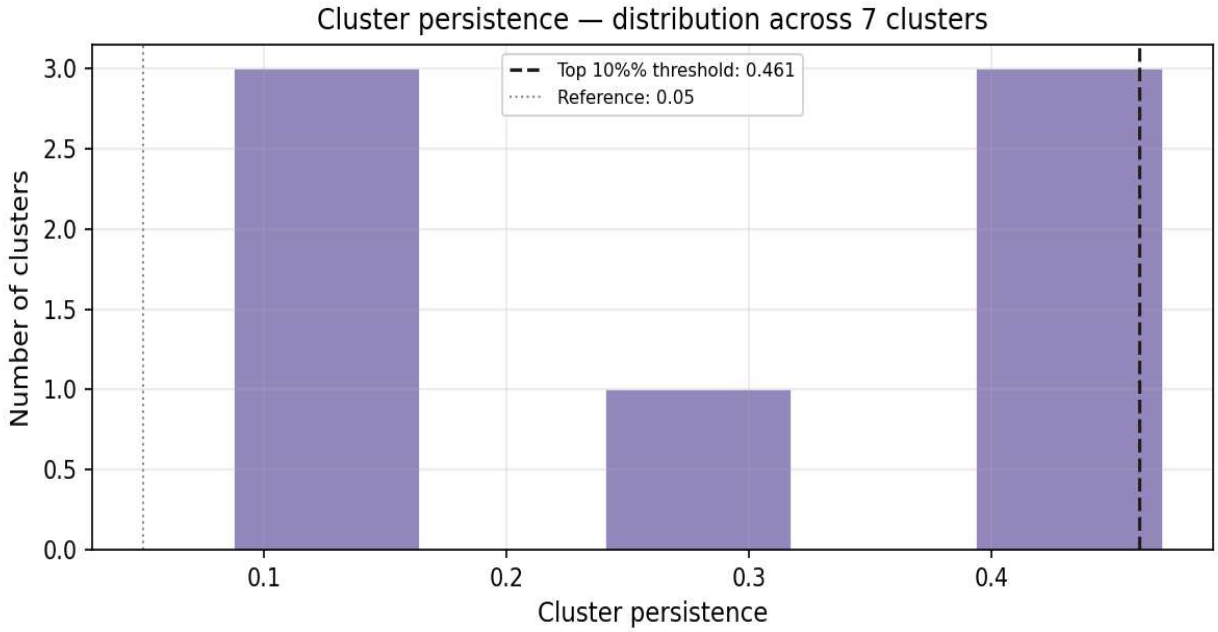
Each histogram shows how a metric is distributed across all clusters. Ideal training clusters sit in the right tail of size, silhouette, and persistence, and the left tail of SSE and noise-neighbour fraction.






Silhouette score — distribution across 7 clusters
Top 10%% threshold: 0.839
Number of clusters
Silhouette score
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Noise-neighbour fraction — distribution across 7 clusters
Bottom 10%% threshold: 0.145
Reference: 0.5
Number of clusters
Noise-neighbour fraction
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.1
0.2
0.3
0.4
0.5


Cluster persistence — distribution across 7 clusters
Top 10%% threshold: 0.461
Reference: 0.05
Number of clusters
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
Cluster persistence

Top-10% Cluster Summary Table

A cluster is included if it falls in the top 10% of ANY per-cluster metric (or bottom 10% for SSE and noise-neighbour fraction where lower is better). Sorted by silhouette score descending. Use this table to identify the best candidate cluster for model training.

Cluster	Size	SSE	Silhouette	Noise nbr frac	Persistence
0	1,189	461.42	0.8700	0.087	0.4703
5	140	3.82	0.8182	0.193	0.3964
2	21,027	198475.16	0.2054	0.286	0.1314